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Bài tập về nhà 14/4/2026

Bài tập 1: Viết chương trình chat-server:

Kết quả: các client nhận được nội dung tin nhắn kèm mốc thời gian

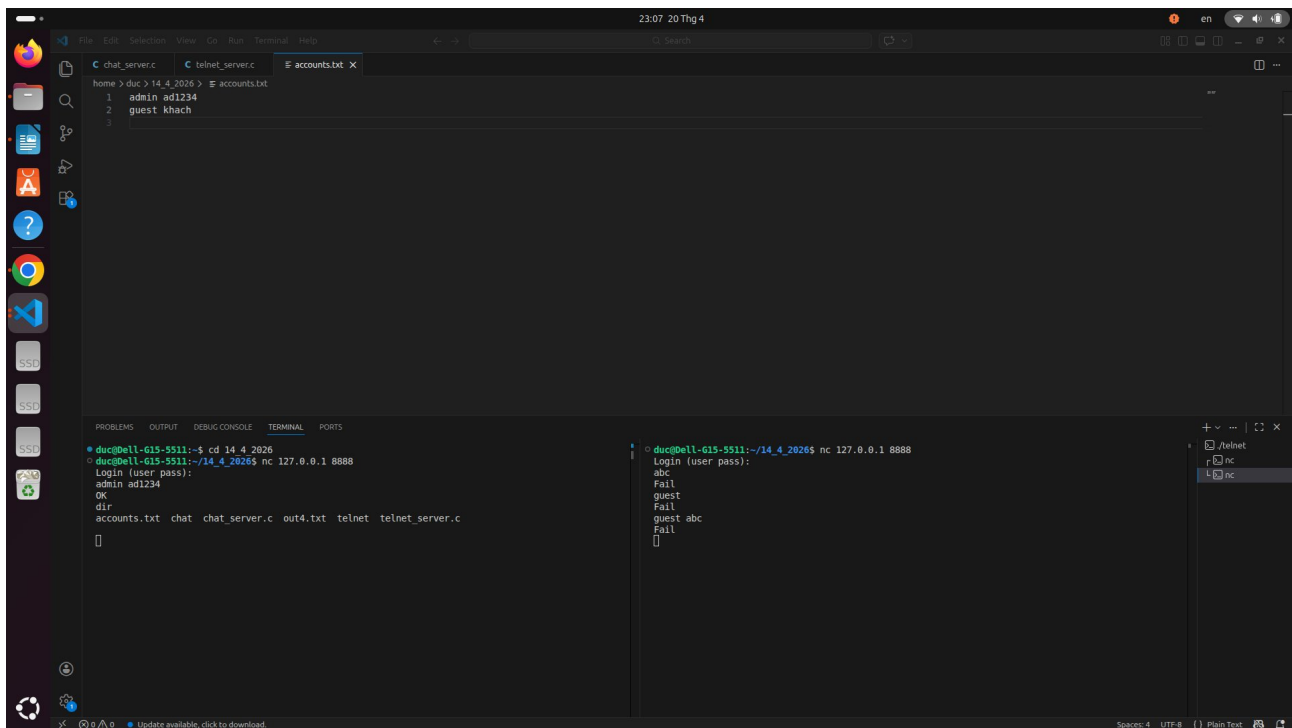
```
22:59 20Thg 4
C chat_server C #include <stdio.h>
32 void send_file(int fd, char *file) {
40     fclose(f);
41 }
42
43 int main() {
44     int listen_fd = socket(AF_INET, SOCK_STREAM, 0);
45
46     struct sockaddr_in addr;
47     addr.sin_family = AF_INET;
48     addr.sin_port = htons(PORT);
49     addr.sin_addr.s_addr = INADDR_ANY;
50
51     bind(listen_fd, (struct sockaddr*)&addr, sizeof(addr));
52     listen(listen_fd, 5);
53
54     struct pollfd fds[MAX_CLIENTS + 1];
55     client clients[MAX_CLIENTS];
56
57     for (int i = 0; i < MAX_CLIENTS; i++) {
58         clients[i].fd = -1;
59         fds[i+1].fd = -1;
60     }
61
62     fds[0].fd = listen_fd;
63     fds[0].events = POLLIN;
64
65     while (1) {
66         int n = poll(fds, MAX_CLIENTS + 1, -1);
67         if (n < 0) continue;
68         if (fds[0].fd < 0) continue;
69         if (fds[0].events & POLLIN) {
70             int new_fd = accept(listen_fd, NULL, NULL);
71             if (new_fd < 0) continue;
72             struct sockaddr_in new_addr;
73             socklen_t len = sizeof(new_addr);
74             int i = 0;
75             while (i < MAX_CLIENTS) {
76                 if (clients[i].fd < 0) {
77                     clients[i].fd = new_fd;
78                     fds[i+1].fd = new_fd;
79                     fds[i+1].events = POLLIN;
80                     break;
81                 }
82                 i++;
83             }
84         }
85         for (int i = 1; i <= MAX_CLIENTS; i++) {
86             if (fds[i].fd < 0) continue;
87             if (fds[i].events & POLLIN) {
88                 int n = read(fds[i].fd, buf, sizeof(buf));
89                 if (n < 0) {
90                     close(fds[i].fd);
91                     clients[i].fd = -1;
92                     fds[i+1].fd = -1;
93                     continue;
94                 }
95                 if (n == 0) {
96                     close(fds[i].fd);
97                     clients[i].fd = -1;
98                     fds[i+1].fd = -1;
99                     continue;
100                }
101                printf("Client %d: %s\n", i, buf);
102                send_file(fds[i].fd, "chat.txt");
103                printf("2026/04/20 22:59:49 xyz: Chao ban\n");
104            }
105        }
106    }
107}
```

```
ducc@bell-615-5511:~$ cd 14.4 2026
ducc@bell-615-5511:~/14.4 2026$ nc 127.0.0.1 8888
Nhap: id: name
abc: Nguyen Van A
OK
Xin chao
2026/04/20 22:59:49 xyz: Chao ban
[]

ducc@bell-615-5511:~/14.4 2026$ nc 127.0.0.1 8888
nc: missing port number
ducc@bell-615-5511:~/14.4 2026$ nc 127.0.0.1 8888
Nhap: id: name
xyz: Tran Van B
OK
2026/04/20 22:59:45 abc: Xin chao
Chao ban
[]
```

Bài tập 2: Viết chương trình telnet-server:

Kết quả: Đúng thông tin đăng nhập thì sẽ được tiếp tục thực hiện các lệnh, khi nhập khai 1 trong 2



The screenshot shows a VS Code editor with three files open: `chat_server.c`, `telnet_server.c`, and `accounts.txt`. The `accounts.txt` file contains the following content:

```
1 admin ad1234
2 guest khach
3
```

The terminal window shows the execution of the `telnet_server.c` program. The user runs `cd 14_4_2026` and then `nc 127.0.0.1 8888`. The program prompts for a login (user pass):

```
Login (user pass):
admin ad1234
OK
dir
accounts.txt chat chat_server.c out4.txt telnet telnet_server.c
[]
```

The terminal also shows the output of the `nc` command, which is:

```
ducdell-615-5511:~/14_4_2026$ nc 127.0.0.1 8888
Login (user pass):
abc
Fail
guest
Fail
guest abc
Fail
[]
```

trường thông tin thì báo lỗi